# Install Manual
# Full Build

How to install, update and look after the ASL-DVS web tools on a Raspberry Pi 3, 4 or 5 running AllStarLink with DVSwitch.

DASHBOARD SYSMON INSTMON WIFIMON 44HELPER

FILES COVERED BY THIS EDITION

| TOOL | FILE | VERSION |
|---|---|---|
| **instmon** | `instmon_v2_43_0_20261012.py` | 2.43.0 |
| **Installer** | `install_all_v7_1_20261012.sh` | 7.1 |
| **Dashboard** | `nodecontrol_v9_3_76_20261012.py` | 9.3.76 |
| **SysMon** | `sysmon_v6_13_79_20261012.py` | 6.13.79 |
| **wifimon** | `wifimon_v5_30_20261012.py` | 5.30 |
| **44helper** | `44helper_v0_0_181_20261012.py` | 0.0.181 |

# In this manual

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## How to read the commands

Commands are typed on the Pi, in an SSH session or on a keyboard and screen. Lines starting with `#` are notes, not commands. Replace `<pi-address>` with the Pi's IP address or host name.

```
# a note about the next line
sudo systemctl status sysmon
```

**NOTE** The scripts on GitHub are comment-stripped copies. Each tool's change history is in the `changelogs/` folder of the repo.

# What gets installed

Each tool is a single file. Each one runs as its own systemd service and serves its own page in the browser.

| TOOL | SERVICE | PORT | WHAT IT DOES |
|---|---|---|---|
| **Dashboard** | `nodecontrol` | 8989 | Node control. Connect ASL and EchoLink nodes; tune DMR, YSF, P25, NXDN, D-STAR, XLX and M17; favorites; Phone. |
| **SysMon** | `sysmon` | 9999 | System monitor and health checks for the node, DVSwitch and its config files. |
| **instmon** | `instmon` | 8990 | Installer and manager. Install, update, start and stop the other tools; GitHub updates; Quiet System. |
| **wifimon** | `wifimon` | 8991 | WiFi and supply-voltage watchdog. Shuts the Pi down cleanly on sustained low voltage or lost network. |
| **44helper** | `44helper` | 9997 | Optional. 44Net Connect, firewall, router and SvxLink helper. |

## The full build

SysMon and the Dashboard come in two builds. The **full build** is for a Pi 3, 4 or 5. Its files have no `_pi02w_` in the name, and its `VERSION` has no `-pi02w` ending.

**DASHBOARD · FULL**

Every tab, including Phone and all digital modes.

- Keeps the AllStar and EchoLink lists in memory for the fastest search.
- Polls Asterisk and refreshes the page at the full rate.

**SYSMON · FULL**

Every tab, including the ones the Pi02w build drops:

- **Console**: the in-browser Asterisk console and the Troubleshooting checks.
- Net, Reg, Firewall and Zello.

instmon, wifimon, 44helper and the installer are **common**: one build for every Pi.

**PI ZERO 2 W?** Use the Pi02w build and its own manual, `docs/ASL-DVS_Pi02w_Install_Manual.pdf`. The full build is too heavy for 512 MB.

**NOTE** The Watchdog script (`asl_dvs_watchdog`) is not needed. The Dashboard's own service already restarts it if it stops, and the installer removes an old Watchdog if it finds one.

# Before you start

## What the Pi needs

- A Raspberry Pi 3, 4 or 5 already running **AllStarLink** with **DVSwitch** (Analog_Bridge and MMDVM_Bridge) set up.
- `python3` and systemd. Both come with Raspberry Pi OS.
- A network connection, and SSH access with `sudo`.

## Set a root password

The web pages ask for a login, and they check it against the Pi's **root** password. Raspberry Pi OS ships with root locked, so set a password once:

```
sudo passwd root
```

# Get the files and install instmon

## Get the files

Put all the files in one folder on the Pi, for example your home folder:

```
# files from the main branch on GitHub
cd ~
B=https://raw.githubusercontent.com/kd8pgk/ASL-DVS/main
wget $B/instmon_v2_43_0_20261012.py
wget $B/install_all_v7_1_20261012.sh
wget $B/nodecontrol_v9_3_76_20261012.py
wget $B/sysmon_v6_13_79_20261012.py
wget $B/wifimon_v5_30_20261012.py
# optional
wget $B/44helper_v0_0_181_20261012.py
```

**TIP** Newer files may be on GitHub by the time you read this. Check the repo's file list and use the newest SysMon and Dashboard files *without* `_pi02w_` in the name.

instmon goes first. Once it is running, it can install and update everything else from the browser, and it updates itself.

**STEP 1 · INSTMON**

```
sudo python3 instmon_v2_43_0_20261012.py --install
```

Then open `http://<pi-address>:8990` and log in as **root**.

## What `--install` does

- Copies the file into place and writes `instmon.service`.
- Writes the shared launcher `/usr/local/bin/asl_dvs_launch.py` (see *How the tools start*, section 8).
- Enables the service so it starts on boot, and starts it now.

## Optional: stage the other files in instmon

Instead of running the installer by hand (Step 2), you can upload the files on instmon's page, or let **Check GitHub** download them into instmon's library, and press **Install** there. The installer also finds files staged in instmon's library.

# Install the rest with the installer

**STEP 2 · INSTALLER**

```
sudo bash install_all_v7_1_20261012.sh
```

## What it asks

1. **Operation.** If tools are already installed: `i` install / reinstall, `u` update binaries only, or `q` quit. Config is kept either way.
2. **Version.** If more than one version of a tool is found, pick one (Enter takes the newest).
3. **First-run identity.** On a fresh node: your callsign, ASL node number and a display label. They go into `/etc/asl_dvs/asl_dvs.conf`.

## Where it looks for files

The installer's own folder, the current folder, `/root`, `/home/pi`, your home folder, `/tmp` and instmon's library. Any subset of the tools can be installed: it installs what it finds.

## Which build it picks

For SysMon and the Dashboard the build is chosen first, then only that build's files are offered:

1. `SYSMON_VARIANT` / `DASH_VARIANT` = `pi02w` or `full`, if set;
2. otherwise the build already installed;
3. otherwise **full on a Pi 3, 4 or 5** (pi02w only on a Pi Zero 2 W).

```
# force the full build, for example to switch back from pi02w
sudo SYSMON_VARIANT=full DASH_VARIANT=full bash install_all_v7_1_20261012.sh
```

**CAREFUL** If no file of the chosen build is found, that tool is skipped with a warning. The other build is never used in its place.

## Without questions

Add `--non-interactive` (or `-y`). It picks the newest files, updates an existing install, and takes the identity from environment variables on a fresh node:

```
sudo AUTO_CALLSIGN=KD8PGK AUTO_NODE=652701 AUTO_LABEL="Travel Node" \
     bash install_all_v7_1_20261012.sh --non-interactive
```

When it finishes, the installer checks that each service is running and listening on its port.

5 · STEP 3

# Open the pages and log in

| PAGE | ADDRESS |
|---|---|
| **Dashboard** | `http://<pi-address>:8989` |
| **SysMon** | `http://<pi-address>:9999` |
| **instmon** | `http://<pi-address>:8990` |
| **wifimon** | `http://<pi-address>:8991` |
| **44helper** | `http://<pi-address>:9997` |

Log in as **root** with the root password from section 2. The pages use plain HTTP, so only open them on your own network or over a VPN.

> **TIP** On the first visit to the Dashboard, check that the callsign and node number in the header are right.

6 · STEP 4

# Set up wifimon

On a first install, wifimon runs with both **watchdogs off**: it shows the WiFi and the voltage, but doesn't reconnect or shut the node down. When you log in to its page (port 8991) it asks *"Turn the watchdogs on?"*. Add your WiFi networks first, then turn them on there or in the **Watchdog** card:

- **WiFi watchdog:** reconnects to saved networks, and shuts the node down after 3 minutes with no connection. It waits, with no timer, until at least one network is saved.
- **Voltage watchdog:** shuts the node down after 30 seconds of low voltage.

Updates and reinstalls keep the watchdogs as they were. Add the networks it may reconnect to:

```
sudo python3 /usr/local/bin/wifimon.py --setup-wifi
```

You can also manage networks and each WiFi adapter on wifimon's page (port 8991).

> **CAREFUL** wifimon shuts the Pi down cleanly on sustained low voltage or a sustained loss of network. Use a good power supply and cable: a Raspberry Pi is sensitive to voltage drop.

# Updating from GitHub with instmon

The **GitHub Updates** card in instmon reads the repo's file list directly.

**CHECK GITHUB**

Shows the newest GitHub copy of each tool and whether it is newer than yours. Full and Pi02w builds get separate rows; rows for the build this Pi isn't using are tagged *other build*.

**UPDATE**

Downloads one file into instmon's library and checks it first: the same file GitHub listed, a version line, the right first line, a syntax check and `--install` support. Nothing installs until you press **Install**.

**FULL UPDATE**

1. Downloads and checks every new file before anything changes.
2. Pauses the web tools.
3. One tool at a time: saves a copy of the installed file, uninstalls it, installs the new one. If the new one fails, it puts the old one back and moves on.
4. Updates instmon last. A timer reinstalls the old instmon if the new one isn't answering after 90 seconds.
5. Ends with a summary.

## Builds and Full Update

On a Pi 3, 4 or 5 each tool keeps the build it has, and Check GitHub only offers newer files of that build. The **Pi Zero 2 W build** choice only appears on a Pi Zero 2 W.

## Quiet System

**Quiet System / Restore** pauses SysMon, the Dashboard and 44helper before a big upload or install. Asterisk, the bridges, Allmon3, SSH and wifimon keep running. Everything comes back by itself after 30 minutes.

# How the tools start

| TOOL | STARTED AS |
|---|---|
| **Dashboard (full)** | `/usr/bin/python3 /usr/local/bin/nodecontrol.py` |
| **SysMon (full)** | `/usr/bin/python3 /usr/local/bin/sysmon.py` |
| **instmon, wifimon, 44helper** | `/usr/bin/python3 /usr/local/bin/asl_dvs_launch.py <tool>` |

## The launcher

The common tools start through `/usr/local/bin/asl_dvs_launch.py`. Started directly, Python compiles the whole file on every start and keeps that memory. The launcher loads the tool as a module instead, so Python saves the compiled copy in `__pycache__` and reuses it. That is about half the memory, and twice as fast to start. The full SysMon and Dashboard are started directly.

## Three more memory savers

Launcher version 2 also turns these on for the tools it starts. Each has an off switch:

| SAVER | WHAT IT DOES | OFF SWITCH |
|---|---|---|
| **-OO** | Drops the built-in help text (docstrings) from the loaded code. | `/etc/asl_dvs/launch_no_optimize` |
| **Pool cap** | `MALLOC_ARENA_MAX=2`: at most 2 memory pools instead of up to 8 per CPU core. | `/etc/asl_dvs/launch_no_arena_cap` |
| **Trim** | 1 minute after start, then every 5 minutes, hands freed memory back to Linux. | `/etc/asl_dvs/launch_no_trim` |

To turn one off, create its file and restart the services. Delete the file and restart to turn it back on:

```
sudo touch /etc/asl_dvs/launch_no_arena_cap
sudo systemctl restart instmon wifimon 44helper
```

**NOTE** The first start after an update compiles the tool once more, so its memory runs high until the next restart.

## Checking memory

```
free -m
# memory of each Python tool, in MB
ps -eo rss,args --sort=-rss | awk '/python3/ && !/awk/ {printf "%6.1f MB  %s\n", $1/1024, substr
($0, index($0,$2))}'
```

# Troubleshooting

## A page doesn't load

```
# is the service running?
systemctl status sysmon --no-pager
# its last 30 log lines
journalctl -u sysmon -n 30 --no-pager
# is anything listening on the port?
ss -tlnp | grep 9999
```

Use the service name and port of the tool in question (table in section 1).

## The login is refused

The pages check the **root** password. Set or reset it with `sudo passwd root`.

## The installer says "No full … build found"

The full-build file for that tool (no `_pi02w_` in the name) isn't in any of the folders it searches. Put it next to the installer and run it again.

## The Pi is slow or runs out of memory

- Run the memory commands in section 8.
- Check that the instmon, wifimon and 44helper lines show `asl_dvs_launch.py`. Started as plain `python3 tool.py`, a tool uses about twice the memory: run its `--install` again.
- Close Cockpit's page when you're done: its two `cockpit-bridge` processes use about 27 MB while it is open.

## A tool misbehaves after an update

Turn the memory savers off one at a time (section 8), starting with the pool cap, and restart. If that doesn't help, reinstall the previous version from instmon's library.

## SysMon's own checks

```
sudo python3 /usr/local/bin/sysmon.py --selftest
```

It runs SysMon's security checks against this Pi and prints PASS / WARN / FAIL for each, without starting the web page. On the page, **Console → Troubleshooting** runs the full set of checks, including the installed Dashboard file and the saved favorites.

10 · REMOVING

## Removing the tools

Remove one tool with its own uninstall:

```
sudo python3 /usr/local/bin/sysmon.py --uninstall
```

Remove every tool except instmon:

```
sudo bash uninstall_all_v1_5.sh
```

This leaves `/etc/asl_dvs/`, instmon's library and `/etc/wifimon/wifimon.conf` in place. Remove instmon last with its own `--uninstall`. The launcher is removed once no service uses it.

11 · QUICK REFERENCE

## Quick reference

| TASK | COMMAND |
|---|---|
| Install instmon | `sudo python3 instmon_v2_43_0_20261012.py --install` |
| Install the rest | `sudo bash install_all_v7_1_20261012.sh` |
| Set the login password | `sudo passwd root` |
| Add WiFi networks | `sudo python3 /usr/local/bin/wifimon.py --setup-wifi` |
| Restart the web tools | `sudo systemctl restart sysmon nodecontrol instmon wifimon` |
| A tool's log | `journalctl -u <service> -n 30 --no-pager` |
| Memory overview | `free -m` · `zramctl` |
| Saver off switch | `sudo touch /etc/asl_dvs/launch_no_<optimize\|arena_cap\|trim>` |
| Remove all but instmon | `sudo bash uninstall_all_v1_5.sh` |

| SERVICE | PORT | CONFIG |
|---|---|---|
| **nodecontrol** | 8989 | `/etc/asl_dvs/asl_dvs.conf` |
| **sysmon** | 9999 | `/etc/sysmon/sysmon.conf` |
| **instmon** | 8990 | `/etc/asl_dvs/` · library in `/etc/asl_dvs/instmon_library` |
| **wifimon** | 8991 | `/etc/wifimon/wifimon.conf` |
| **44helper** | 9997 | `/etc/44helper/44helper.conf` |